1. A healthcare startup wants to build a model to predict the onset of diabetes. The dataset has

50 features, but the team wants to use only the most relevant ones. How can they apply

feature selection effectively?

- Collect the data
- Preprocess the data for missing values.
- Using correlation method remove the columns which are highly correlated.
- Split the train and test data.
- Apply selectKbest or RFE (with Logistic Regression Algorithm since it is classification

problem) to reduce the features with relevant ones.

- Train the model using selected features
- Evaluate Accuracy for the model.

2. You're creating a new project for managing student data. The project name is student_portal.

What is the step-by-step command to start it?write logic for this django project

- Open the command prompt
- Go to the path where the project to be
- django-admin startproject student_portal
- student_portal folder is created in the path.

3. A company is analyzing customer purchase patterns with 200+ behavioral features. How can they

reduce dimensionality without losing predictive power?

- Collect the data
- Preprocess the data for missing values.
- Use PCA for dimensionality reduction.
- Train the model using top features
- Evaluate Accuracy or RMSE for the model.

4.A digital library wants to recommend books to readers based on what similar readers liked. How

should they design this system?

Collaborative recommendation system. (User or item based)

- Create the pivot table with user and the books Id and the rating of the books.
- Find the cosine similarity score between the users or the between the items.

• Create prediction table if user reads the books.

• List the similar users read the books.

• Remove the books from the user list if user already read the books.

• For the remaining books, select the books which has more rating in the prediction table.

• Recommend those books to the users.

5. A bank wants to assess the risk level of credit applicants using only the most important financial

indicators. How can they reduce the number of features?

• Collect the data

• Preprocess the data for missing values.

• Using correlation method remove the columns which are highly correlated.

• Split the train and test data.

• Apply RFE (with Logistic Regression Algorithm since it is classification problem) to

reduce the features with relevant ones.

• Train the model using selected features

• Evaluate Accuracy for the model.

6. A news app wants to recommend articles based on both article similarity and user reading history.

How can they implement a hybrid system?

Content Based Recommendation

• Create the pivot table for user and the items.

• Create a vector list for whole user with to item vector

• Create the content vector using user and item vectors

• Evaluate the content vector using rating

7. You're building a spam detection model and have thousands of text features from emails. How do

you identify the most useful ones?

• Collect the data

• Preprocess the data for missing values.

• Using correlation method remove the columns which are highly correlated.

• Split the train and test data.

• Apply RFE (with Logistic Regression Algorithm since it is classification problem) to

reduce the features with relevant ones.

• Train the model using selected features

• Evaluate Accuracy for the model.

8. An ed-tech platform wants to recommend courses based on what similar learners have enrolled in.

What steps would you take?

Collaborative recommendation system. (User or item based)

• Create the pivot table with user and the course Id and the rating of the course.

• Find the cosine similarity score between the users or the between the items.

• Create prediction table if user learn the course.

• List the similar users learn the course.

• Remove the course from the user list if user already learn the course.

• For the remaining courses, select the course which has more rating in the prediction

table.

• Recommend those course to the users.

9.You’re developing a car price prediction tool. With 100+ features (e.g., brand, mileage, engine

type), how do you reduce complexity?

• Collect the data

• Preprocess the data for missing values.

• Using correlation method remove the columns which are highly correlated.

• Split the train and test data.

• Apply selectKbest to reduce no of the features with relevant ones.

• Train the model using selected features

• Evaluate Accuracy for the model.

10. How do you recommend products to new users who haven’t interacted with anything yet?

Popularity based recommendation

• Create the pivot table for user and the items.

• Create a vector list for whole user with to item vector

• Create the content vector using user and item vectors

• From the content vectors select the top 5 predicted values. For the top5 values select the

products.

1. A retail chain wants to forecast product sales using factors like holidays, promotions,

location, and inventory. The dataset has 80 features. How should they select the most impactful ones?

- Collect the data
- Preprocess the data for missing values.
- Using correlation method remove the columns which are highly correlated.
- Split the train and test data.
- Apply RFE (with Linear Regression Algorithm) to reduce the features with relevant ones.
- Train the model using selected features
- Evaluate R2score for the model.

2.A podcast platform wants to recommend new podcasts to users based on the categories and hosts

they follow. How should they design this content-based system?

- Create the pivot table for user and the items.
- Create a vector list for whole user with to item vector (similarity matrix)
- Create the content vector using user and item vectors
- Select the categories and hosts users follow.
- From the content vectors select the select the podcasts belong to that categories and hosts and that user does not know.
- Recommend that podcasts.

3.An industrial firm wants to predict machine failure using sensor readings and system logs (200+

features). How do they choose which inputs to use?

- Collect the data
- Preprocess the data for missing values.
- Using correlation method remove the columns which are highly correlated.
- Split the train and test data.
- Apply RFE (with Random Forest Algorithm) to reduce the features with relevant ones.
- Train the model using selected features
- Evaluate R2score for the model.

4. A fashion app wants to suggest outfits to users based on their previous likes, color preferences,

and seasonal trends. How should they implement the recommendation?

• Create the pivot table for user and the items.

• Create a item similarity matrix

• Create the prediction table for the items and user

• From recommendation list selecting only highest rated(predicted) value

5.A research lab wants to predict heart disease using patient data. The dataset has lab results, habits,

and vitals. How should they select only the most relevant features?

• Collect the data

• Preprocess the data for missing values.

• Using correlation method remove the columns which are highly correlated.

• Split the train and test data.

• Apply selectkbest for classification to reduce the features with relevant ones.

• Train the model using selected features

• Evaluate Accuracy score for the model.

6. A job portal wants to recommend job listings based on a user’s resume and previous applications.

How can they build a recommendation system?

• Create the pivot table for user and the items.

• Create a user similarity matrix with the job description

• Create the prediction table for the user and items

• From recommendation list selecting only highest rated(predicted) value

7. A smart home system wants to predict energy usage using readings from multiple sensors. With

100+ features, how do they reduce complexity?

• Collect the data

• Preprocess the data for missing values.

• Using correlation method remove the columns which are highly correlated.

• Split the train and test data.

• Apply selectkbest or RFE (with Linear Regression Algorithm) to reduce the features

with relevant ones.

• Train the model using selected features

• Evaluate R2score for the model.

8. An e-learning platform wants to suggest tutorials based on a learner's skill gaps identified from

quizzes. How should they build this system?

• Create the pivot table for learner and the tutorials.

• Create a vector list for whole users with respect to tutorials vector (similarity matrix)

• Create the content vector using user and item vectors

• From the content vectors select the tutorials that the learner has gap.

• Recommend those tutorials.

9.A social media app wants to predict post engagement (likes/comments) using post features, timing,

and user activity. How do they select the right features?

• Collect the data

• Preprocess the data for missing values.

• Using correlation method remove the columns which are highly correlated.

• Split the train and test data.

• Apply selectkbest/RFE/any feature selection (Logistic algorithm or any best

algorithm for classification) to reduce the features with relevant ones.

• Train the model using selected features

• Evaluate Accuracy score for the model.

10. Write code logic for this:

Now you want to create an app named attendance in the project and register it.write logic for this

Django project

• Go to your Particular Folder

• python manage.py attendance

• add the attendance project in the settings.py